Seaborn Complete Compact Handbook

Beginner + Intermediate Notes

What is Seaborn?

Seaborn is a statistical visualization library built on top of Matplotlib. It simplifies creating attractive and informative visualizations while integrating well with Pandas DataFrames.

```
import seaborn as sns
```

Why Seaborn?

Provides better default styling, statistical plots, less code, and easier exploratory data analysis than Matplotlib.

```
sns.lineplot(data=df,x='Month',y='Sales')
```

Built-in Datasets

Seaborn provides datasets such as tips, iris, titanic, diamonds, flights and penguins for practice.

```
tips = sns.load_dataset('tips')
```

Themes & Styles

Themes control the appearance of plots. Common styles: whitegrid, darkgrid, white, dark and ticks.

```
sns.set_style('whitegrid')
```

scatterplot()

Shows relationship between two variables. Each dot represents one observation.

```
sns.scatterplot(data=tips,x='total_bill',y='tip')
```

lineplot()

Used for trends and ordered data. Common in sales and time-series analysis.

```
sns.lineplot(data=df,x='Month',y='Sales',marker='o')
```

histplot()

Shows distribution of a numerical variable using bins.

```
sns.histplot(data=tips,x='total_bill',kde=True)
```

kdeplot()

Smooth estimate of data distribution. Think of it as a smoothed histogram.

```
sns.kdeplot(data=tips,x='total_bill',fill=True)
```

displot()

Figure-level distribution plotting function supporting histograms and KDE.

```
sns.displot(data=tips,x='total_bill',kde=True)
```

barplot()

Displays aggregated statistics (mean by default) across categories.

```
sns.barplot(data=tips,x='day',y='total_bill')
```

countplot()

Displays frequency/count of categories.

```
sns.countplot(data=tips,x='sex')
```

boxplot()

Shows median, quartiles, IQR and outliers. Important for ML and data cleaning.

```
sns.boxplot(data=tips,x='day',y='total_bill')
```

violinplot()

Combines boxplot and KDE information into one plot.

```
sns.violinplot(data=tips,x='day',y='total_bill')
```

stripplot()

Shows every observation. Useful for inspecting raw data.

```
sns.stripplot(data=tips,x='day',y='total_bill',jitter=True)
```

swarmplot()

Smart stripplot that avoids overlapping points.

```
sns.swarmplot(data=tips,x='day',y='total_bill')
```

pointplot()

Displays means/statistics using points instead of bars.

```
sns.pointplot(data=tips,x='day',y='tip')
```

regplot()

Scatterplot plus regression line for relationship analysis.

```
sns.regplot(data=tips,x='total_bill',y='tip')
```

lmplot()

Advanced regression plotting with hue, row and col grouping.

```
sns.lmplot(data=tips,x='total_bill',y='tip',hue='sex')
```

heatmap()

Represents values using colors. Commonly used for correlation matrices.

```
sns.heatmap(df.corr(numeric_only=True),annot=True)
```

clustermap()

Heatmap with clustering and dendrograms. Requires SciPy.

```
sns.clustermap(df.corr(numeric_only=True))
```

pairplot()

Automatically visualizes relationships among multiple variables.

```
sns.pairplot(iris,hue='species')
```

jointplot()

Shows relationship and distributions of two variables together.

```
sns.jointplot(data=tips,x='total_bill',y='tip',kind='reg')
```